

CORTEX: Self-Evolving Cognitive Memory for Autonomous AI Agents

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Abstract. Persistent memory remains a fundamental limitation of large language model (LLM)-based autonomous agents. Current memory systems for AI agents typically support a single memory type with flat retrieval, lacking the multi-layered cognitive structure that enables biological intelligence to consolidate, forget, self-correct, and evolve retrieval strategies over time. We present **CORTEX** (Cognitive Orchestrated Retrieval with Temporal EXperience), a nine-layer cognitive memory architecture that unifies four biologically-inspired memory types—working, episodic, semantic, and procedural—with a hippocampal hybrid search index, an Ebbinghaus forgetting curve, a sleep/dream consolidation cycle, a self-improvement engine, a genetic algorithm for retrieval strategy evolution, a meta-cognition layer with active gap filling, prompt-aware cognitive assembly that automatically selects prompt engineering techniques based on task complexity, and bidirectional synchronization with Obsidian knowledge vaults. CORTEX operates fully offline with zero external API dependencies, persists all state in SQLite, and requires only Python and NumPy. A multi-backend embedding system with automatic detection enables optional semantic embeddings via sentence-transformers or ONNX Runtime while preserving zero-mandatory-dependency operation. Experimental evaluation on 179 unit tests demonstrates correctness across all subsystems, with sub-millisecond memory operations and convergent evolution of retrieval strategies within 10 generations. To our knowledge, CORTEX is the first architecture to combine all nine cognitive layers—including self-evolution via genetic algorithms, active knowledge gap filling, and automatic prompt engineering—in a single, embeddable memory system for autonomous agents.

Keywords: Cognitive memory architecture · Autonomous agents · Self-evolving AI · Knowledge management · Memory consolidation · Genetic algorithms

1 Introduction

The emergence of LLM-based autonomous agents [26,23] has created an urgent need for persistent, structured memory systems that go beyond simple vector stores. While human cognition benefits from multiple specialized memory

systems—working memory for immediate context, episodic memory for experiences, semantic memory for facts, and procedural memory for skills [24,21,4]—most current AI agent memory systems implement only one or two of these types, with no mechanism for self-improvement or evolution.

Current approaches such as Mem0 [15], MemGPT/Letta [17], and Zep [28] provide valuable memory persistence but suffer from three key limitations: (1) they lack multi-type memory organization inspired by cognitive science, (2) they have no self-evolution mechanism for retrieval strategies, and (3) they do not actively detect and fill knowledge gaps. Furthermore, no existing system provides bidirectional synchronization with personal knowledge management tools, creating a disconnect between agent memory and human-curated knowledge bases.

Contributions. We introduce CORTEX, a nine-layer cognitive memory architecture that addresses these gaps:

1. **Unified multi-type memory** with four biologically-inspired stores (working, episodic, semantic, procedural) connected through a hippocampal hybrid search index combining vector similarity, BM25, temporal, graph, and importance signals.
2. **Cognitive lifecycle management** via an Ebbinghaus forgetting curve ($R = e^{-t/S}$) and a sleep/dream consolidation cycle that promotes memories across types.
3. **Self-evolution** through a genetic algorithm that evolves retrieval strategy parameters using tournament selection with elitism, producing measurable fitness convergence.
4. **Active knowledge gap filling** with a four-class taxonomy (inferrable, searchable, askable, consolidatable) and five auto-fill strategies.
5. **Bidirectional Obsidian vault synchronization** supporting PARA method detection, wikilink graph mapping, dead link gap detection, and auto-generated Maps of Content.
6. **Automatic prompt engineering** via a PromptAssembler that analyzes task complexity (simple, moderate, complex, knowledge gap, skill execution) and selects optimal combinations from ten prompt techniques including chain-of-thought, ReAct, few-shot, and self-critique, with outcome recording for self-improvement.
7. **Zero-mandatory-dependency offline operation** using SQLite with WAL mode and a multi-backend embedding system that auto-detects the best available provider (ONNX Runtime, sentence-transformers, or hash-based TF-IDF), enabling real semantic embeddings without requiring external APIs.

2 Related Work

2.1 Memory Systems for AI Agents

Mem0 [15] provides a memory layer that automatically extracts and stores facts from conversations, supporting graph-based relationships. However, it

lacks multi-type memory organization and self-evolution. **MemGPT/Letta** [17] treats LLMs as operating systems with virtual context management, implementing main and archival memory tiers. It does not support procedural memory or forgetting curves. **Zep** [28] offers long-term memory with temporal awareness and entity extraction but lacks consolidation cycles and self-improvement. **Hindsight** [11] implements self-improving memory through reflection but does not incorporate genetic evolution or gap filling. **Cognee** [5] provides graph-based RAG with deterministic outputs but focuses on retrieval rather than cognitive memory management.

Recent surveys [29,10] identify memory management as a critical open problem, with no existing system combining all four cognitive memory types with self-evolution.

2.2 Cognitive Architectures

Classical cognitive architectures provide theoretical foundations. **ACT-R** [2] models human cognition with declarative and procedural modules, inspiring our dual long-term memory design. **SOAR** [13] implements production rules and chunking, analogous to our procedural memory with reliability tracking. **Global Workspace Theory** [3] posits a shared workspace for competing cognitive processes, reflected in our hippocampal index that arbitrates across memory types. Park et al. [18] demonstrated generative agents with reflection and planning using memory streams, while Summers et al. [23] formalized cognitive architectures for language agents.

2.3 Knowledge Management Methods

The **Second Brain** methodology [8] and the **PARA** method (Projects, Areas, Resources, Archive) provide organizational frameworks that CORTEX detects and maps to memory types during Obsidian vault ingestion. The **Zettelkasten** method [1] emphasizes atomic, interlinked notes—a structure our wikilink graph parser naturally captures.

2.4 Agent Frameworks

Modern agent frameworks including AutoGen [27], CrewAI [6], LangGraph [14], and OpenClaw [16] provide orchestration for multi-agent systems but delegate memory management to external stores. CORTEX is designed as an embeddable memory engine for any such framework.

2.5 Self-Improving AI Systems

Voyager [25] maintains a skill library that grows through exploration, similar to our procedural memory. Reflexion [20] uses verbal reinforcement learning from mistakes, paralleling our self-improvement engine. MemoryBank [12] applies forgetting curves to LLM memory but lacks multi-type organization and evolution.

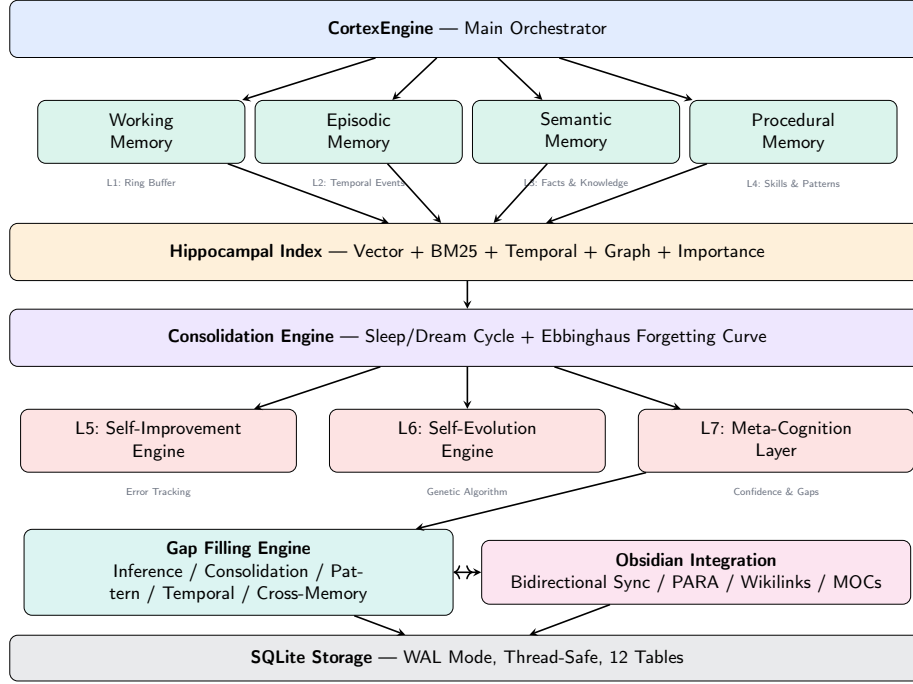


Fig. 1. CORTEX nine-layer cognitive memory architecture. Layers L1–L4 provide multi-type storage; the hippocampal index unifies retrieval; the consolidation engine manages memory lifecycle; layers L5–L7 provide self-awareness; gap filling and Obsidian integration connect to external knowledge.

Neural Turing Machines [9] and End-to-End Memory Networks [22] provide differentiable memory access but operate at the neural network level rather than the agent architecture level.

3 CORTEX Architecture

CORTEX comprises nine interconnected layers orchestrated by a central `CortexEngine` class. Figure 1 presents the overall architecture.

3.1 Layer 1: Working Memory

Working memory implements a fixed-capacity ring buffer (class `WorkingMemory`, default capacity $C = 20$) that maintains the agent’s immediate context. When the buffer is full, the oldest item is evicted via FIFO. Each `WorkingItem` stores content, metadata, a creation timestamp, and a slot number. The `push`, `get_recent(n)`, `search(query)`, and `clear()` operations execute in $O(1)$, $O(n)$, $O(C)$, and $O(C)$ time respectively. During consolidation, all working memory items are flushed to episodic memory.

3.2 Layer 2: Episodic Memory

Episodic memory (class `EpisodicMemory`) stores timestamped events as `Episode` dataclasses with fields: `id`, `content`, `embedding` (256-dimensional hash vector), `importance` $\in [0, 1]$, `access_count`, `last_accessed`, `tags`, and `source`. Retrieval combines cosine similarity with retention scoring:

$$\text{score}(e, q) = \text{sim}(\mathbf{v}_e, \mathbf{v}_q) \cdot R(e) \cdot (0.5 + \text{importance}(e)) \quad (1)$$

where $R(e)$ is the Ebbinghaus retention (Section 4.1). Episodes below a minimum retention threshold are excluded from retrieval. Access timestamps are updated on every retrieval (spaced repetition effect).

3.3 Layer 3: Semantic Memory

Semantic memory (class `SemanticMemory`) stores `Fact` objects with additional fields: `category`, `confidence` $\in [0, 1]$, and optional filtering by category and minimum confidence. The scoring function is:

$$\text{score}(f, q) = \text{sim}(\mathbf{v}_f, \mathbf{v}_q) \cdot \text{confidence}(f) \cdot (0.5 + \text{importance}(f)) \quad (2)$$

Categories enable topic-organized retrieval, while the confidence field allows uncertain or inferred knowledge to coexist with verified facts.

3.4 Layer 4: Procedural Memory

Procedural memory (class `ProceduralMemory`) stores `Skill` objects with `pattern`, `trigger`, `success_count`, and `failure_count` fields. Reliability uses Laplace-smoothed Bayesian estimation:

$$\text{reliability}(s) = \frac{s_{\text{success}} + 1}{s_{\text{success}} + s_{\text{failure}} + 2} \quad (3)$$

Retrieval scoring weights similarity by reliability:

$$\text{score}(s, q) = \text{sim}(\mathbf{v}_s, \mathbf{v}_q) \cdot \text{reliability}(s) \cdot (0.5 + \text{importance}(s)) \quad (4)$$

The `record_outcome(skill_id, success)` method tracks real-world performance. The `get_by_trigger(query)` method enables automatic skill selection via trigger matching.

3.5 Hippocampal Index: Hybrid Search

The `HippocampalIndex` class unifies retrieval across all three persistent memory types by combining five signals into a single relevance score (Figure 2):

$$S(m, q) = \sum_{k \in \mathcal{K}} w_k \cdot s_k(m, q) \quad (5)$$

where $\mathcal{K} = \{\text{similarity, bm25, temporal, graph, importance}\}$ and the weights w_k are controlled by the evolved retrieval strategy (Section 6).

The individual signals are computed as follows:

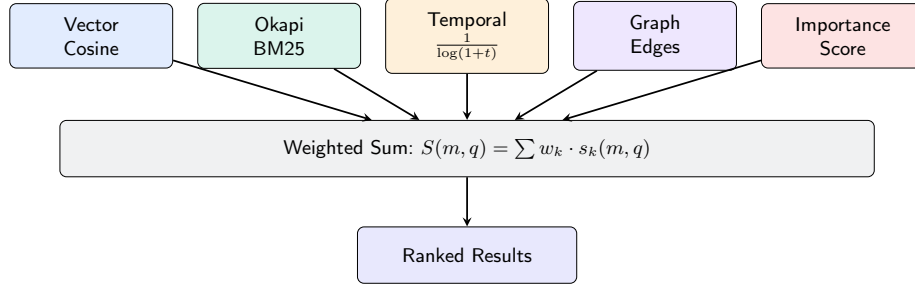


Fig. 2. Hippocampal hybrid search pipeline. Five signals are weighted by evolved strategy parameters and combined into a single relevance score.

- **Vector similarity:** Cosine similarity between query and memory TF-IDF hash embeddings (256-d).
- **BM25:** Okapi BM25 with $k_1 = 1.5$, $b = 0.75$, IDF computed across all memory tables.
- **Temporal:** Recency score $s_{\text{temp}} = 1/\log(1 + t_h)$ where t_h is hours since last access.
- **Graph:** Normalized edge count $s_{\text{graph}} = \min(1, |E_m|/5)$ from the `memory_edges` table.
- **Importance:** Stored importance score directly.

The embedding module supports three backends with automatic detection. The default **hash-based TF-IDF** backend (256-d) requires no additional dependencies: each token is hashed via SHA-256 to a deterministic unit vector, then weighted by $\text{TF} \times 1/\log(1 + \text{count})$ and L2-normalized. For higher-quality semantic matching, CORTEX auto-detects and uses either **ONNX Runtime** (~ 80 MB, no PyTorch dependency) or **sentence-transformers** to provide 384-dimensional contextual embeddings from the `all-MiniLM-L6-v2` model [19]. The detection order is ONNX $\rightarrow$ sentence-transformers $\rightarrow$ hash, ensuring the best available backend is used transparently. An LRU embedding cache avoids redundant computation across repeated queries.

4 Memory Consolidation and Forgetting

4.1 Ebbinghaus Forgetting Curve

Memory retention follows the Ebbinghaus exponential decay model [7]:

$$R(t, S) = e^{-t/S} \quad (6)$$

where $R \in [0, 1]$ is retention, t is elapsed time in hours since last access, and S is the stability parameter. Stability increases with access frequency and importance:

$$S = S_{\text{base}} \cdot (1 + \ln(1 + n_{\text{access}})) \cdot (0.5 + \text{importance}) \quad (7)$$

Algorithm 1: Sleep/Dream Consolidation Cycle

Input: Working memory $\mathcal{W}$, Episodic memory $\mathcal{E}$, Semantic memory $\mathcal{S}$,
Procedural memory $\mathcal{P}$
Output: ConsolidationReport

// Phase 1: Flush working memory

```

1 foreach  $w \in \mathcal{W}.clear()$  do
2    $\mathcal{E}.store(w.content, importance = 0.4, source = consolidation)$ 

// Phase 2: Promote episodic to semantic/procedural
3 foreach  $e \in \mathcal{E}.get\_all()$  do
4   if  $e.access\_count \geq 3$  or  $e.importance \geq 0.7$  then
5     if  $\neg \exists f \in \mathcal{S} : f.content = e.content$  then
6        $\mathcal{S}.store(e.content, confidence = 0.9)$ 
7     if  $LOOKSPROCEDURAL(e.content)$  then
8        $\mathcal{P}.store(e.content, pattern = auto)$ 

// Phase 3: Prune forgotten memories
9 foreach  $e \in \mathcal{E}$  do
10   if  $R(e.last\_accessed, e.access\_count, e.importance) < \tau_{prune}$  then
11      $\mathcal{E}.delete(e.id)$ 
12 Prune semantic memories with confidence  $< 0.2$  and  $access\_count < 2$ 

```

where $S_{base} = 24$ hours. This formulation ensures that frequently accessed, high-importance memories decay slowly, while rarely accessed memories are pruned. A memory is marked for pruning when $R < \tau_{prune} = 0.05$.

4.2 Sleep/Dream Consolidation Cycle

The **ConsolidationEngine** implements a three-phase consolidation cycle (Algorithm 1), modeled after biological memory consolidation during sleep.

Procedural detection uses three regex patterns: conditional actions (“when X , do Y ”), sequential steps (“how to ... first ... then”), and constraints (“always ... before ...”).

The consolidation also supports fragment merging via **consolidate_fragments(topic)**, which deduplicates sentences across multiple episodic memories about the same topic and creates a unified semantic summary with confidence 0.85.

5 Self-Improvement Engine

The **SelfImprovementEngine** (Layer 5) tracks errors, learns correction patterns, and measures learning velocity.

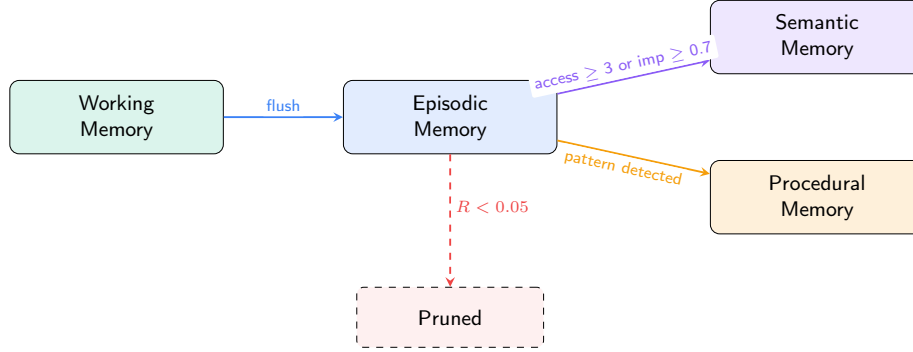


Fig. 3. Memory consolidation flow. Working memory is flushed to episodic; high-value episodes are promoted to semantic or procedural; low-retention memories are pruned.

5.1 Error Tracking and Correction Learning

Each error is logged as an **Error** dataclass with fields: `id`, `description`, `context`, `category`, `severity` $\in [0, 1]$, and `resolved` status. When a correction is logged via `log_correction(correct, error_id, wrong, pattern)`, the engine:

1. Creates a **Correction** record linking the error to the fix.
2. Marks the original error as resolved.
3. Optionally creates a procedural skill from the correction pattern.

Correction matching uses keyword overlap scoring: for a new error description d , the engine searches all corrections and returns the one maximizing $\sum_{w \in \text{tokens}(d)} \mathbb{1}[w \in \text{pattern} \cup \text{wrong}]$.

5.2 Learning Velocity

Learning velocity quantifies the rate of error resolution over time:

$$v_{\text{learn}} = \frac{\Delta P}{\Delta t} = \frac{|\{e \in \mathcal{E}_{\text{err}} : e.\text{resolved} \wedge e.\text{created_at} \geq t_0\}|}{T} \quad (8)$$

where T is the measurement window (default: 7 days) and $t_0 = t_{\text{now}} - T$. The improvement trend is classified as *improving*, *stable*, or *declining* by comparing resolution counts between the current and previous windows.

The **ImprovementReport** dataclass aggregates: total errors, resolved errors, resolution rate, learning velocity, top error categories, correction patterns (sorted by application frequency), and improvement trend.

6 Self-Evolution via Genetic Algorithm

The **EvolutionEngine** (Layer 6) applies a genetic algorithm to evolve retrieval strategy parameters, enabling the memory system to adapt its behavior based on user feedback.

6.1 Strategy Gene Representation

Each strategy θ is a vector of seven tunable parameters:

$$\theta = (w_r, w_f, w_i, w_s, \lambda_d, \tau_c, w_x) \quad (9)$$

where w_r is recency weight $\in [0, 2]$, w_f is frequency weight $\in [0, 2]$, w_i is importance weight $\in [0, 2]$, w_s is similarity weight $\in [0, 3]$, λ_d is decay rate $\in [0.001, 0.1]$, τ_c is consolidation threshold $\in [0.3, 0.9]$, and w_x is cross-memory weight $\in [0, 1.5]$.

6.2 Fitness Function

Fitness is computed from accumulated retrieval feedback:

$$F(\theta) = \frac{\sum_{i=1}^N \mathbb{1}[\text{useful}_i]}{N} \quad (10)$$

where N is the total number of feedback events associated with strategy θ .

6.3 Evolutionary Operators

Selection. Tournament selection with $k = 3$ contestants:

$$\text{parent} = \arg \max_{\theta \in \text{sample}(\Pi, k)} F(\theta) \quad (11)$$

Crossover. Uniform crossover with rate $p_c = 0.6$:

$$\theta_{\text{child}}^{(j)} = \begin{cases} \theta_{\text{parent}_1}^{(j)} & \text{with probability } 0.5 \\ \theta_{\text{parent}_2}^{(j)} & \text{otherwise} \end{cases} \quad (12)$$

Mutation. Gaussian perturbation with rate $p_m = 0.2$ and per-gene probability 0.4:

$$\theta'^{(j)} = \text{clip} \left(\theta^{(j)} + \mathcal{N}(0, \sigma^2(h_j - l_j)^2), l_j, h_j \right) \quad (13)$$

where $\sigma = 0.15$ and $[l_j, h_j]$ are the parameter bounds.

Elitism. The top $E = 2$ strategies survive unchanged to the next generation.

Algorithm 2 formalizes the complete evolution step.

7 Meta-Cognition and Gap Filling

7.1 Confidence Scoring

The **MetaCognitionEngine** (Layer 7) assesses confidence about any topic by aggregating evidence across all memory types:

$$C(m) = 0.35 \cdot C_{\text{qty}} + 0.30 \cdot C_{\text{qlty}} + 0.20 \cdot C_{\text{div}} + 0.15 \cdot \bar{I} \quad (14)$$

where:

- $C_{\text{qty}} = \min(1, \ln(1 + n)/\ln(21))$ is the quantity score (log-scaled memory count).
- $C_{\text{qlty}} = \overline{\text{confidence}}$ is the mean semantic confidence of matching facts.
- $C_{\text{div}} = |\text{sources}|/3$ measures diversity across memory types.
- $\bar{I}$ is the mean importance across all matching memories.

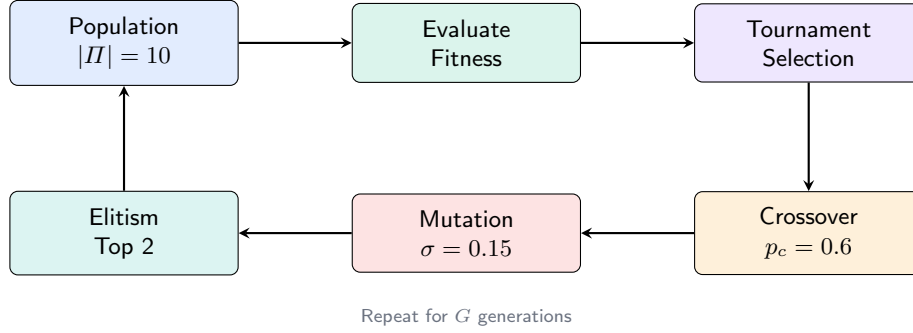


Fig. 4. Genetic algorithm evolution cycle for retrieval strategy optimization.

7.2 Knowledge Gap Detection

The engine detects four types of knowledge gaps:

1. **Sparse topics:** Semantic categories with fewer than 3 memories.
2. **Broken inference chains:** Graph edges pointing to non-existent memories.
3. **Temporal gaps:** Periods exceeding 72 hours with no episodic memories.
4. **Mentioned unknowns:** Tags appearing ≥ 2 times in episodes without semantic coverage.

7.3 Gap Classification and Filling

The `GapFillingEngine` classifies each gap and applies targeted fill strategies (Figure 5):

The five fill strategies are:

1. **Inference:** Combines content from related semantic and episodic memories, deduplicates sentences, stores result with confidence $c = \min(0.8, 0.3 + 0.1 \cdot |\text{sources}|)$.
2. **Consolidation:** Merges episodic fragments about the same topic into a semantic summary via `consolidate_fragments()`.
3. **Pattern:** Applies the best-matching procedural skill, with confidence scaled by reliability.
4. **Temporal interpolation:** Creates placeholder episodic entries for time gaps between known events.
5. **Cross-memory:** Uses any memory type as source to fill gaps in another.

Gap priority is computed as:

$$P = \frac{I \cdot F_q}{A \cdot E_c} \quad (15)$$

where I is the maximum importance of related memories, F_q is the frequency of related queries (minimum 1), A is the gap age in days (minimum 1), and E_c is the existing coverage (minimum 0.1). Priority is clamped to $[0, 1]$.

Algorithm 3 formalizes the gap detection and filling process.

Algorithm 2: Genetic Evolution Step

Input: Population Π , population size $N = 10$, elite count $E = 2$
Output: New population Π'

```

1 Sort  $\Pi$  by fitness descending
2  $\Pi' \leftarrow \Pi[0 : E]$  // Elitism
3 while  $|\Pi'| < N$  do
4   if  $\text{rand}() < p_c$  then
5      $\theta_a \leftarrow \text{TOURNAMENTSELECT}(\Pi, k = 3)$ 
6      $\theta_b \leftarrow \text{TOURNAMENTSELECT}(\Pi, k = 3)$ 
7      $\theta_{\text{child}} \leftarrow \text{UNIFORMCROSSOVER}(\theta_a, \theta_b)$ 
8   else
9      $\theta_{\text{child}} \leftarrow \text{TOURNAMENTSELECT}(\Pi, k = 3)$ 
10  if  $\text{rand}() < p_m$  then
11     $\theta_{\text{child}} \leftarrow \text{GAUSSIANMUTATE}(\theta_{\text{child}}, \sigma = 0.15)$ 
12   $\Pi'.\text{append}(\theta_{\text{child}})$ 
13 Deactivate  $\Pi$ ; persist  $\Pi'$ 
14 return  $\Pi'$ 

```

7.4 Coverage Map

The `get_coverage_map()` method builds a `CoverageMap` by clustering memories by semantic category and frequent tokens, computing coverage as $\min(1, n/10)$ for each cluster. Topics with coverage ≥ 0.7 are classified as well-covered; those below 0.3 as sparse. Temporal coverage groups episodic memories by month.

7.5 Self-Evaluation

The `self_evaluate()` method produces a `SelfEvaluation` with overall quality:

$$Q = 0.25 \cdot \bar{C}_{\text{conf}} + 0.20 \cdot \bar{I} + 0.20 \cdot C_{\text{cov}} + 0.15 \cdot D + 0.20 \cdot R \quad (16)$$

where $\bar{C}_{\text{conf}}$ is mean confidence, $\bar{I}$ is mean importance, C_{cov} is coverage score, $D = |\text{active types}|/3$ is diversity, and R is the fraction of memories accessed in the last 7 days.

8 Prompt-Aware Cognitive Assembly

While CORTEX’s memory layers handle *what* is stored and retrieved, the PromptAssembler addresses *how* retrieved knowledge is presented. This module automatically selects and combines prompt engineering techniques based on task complexity analysis, making CORTEX “prompt-aware”—it does not merely retrieve memories but constructs optimally structured prompts for downstream LLM consumption.

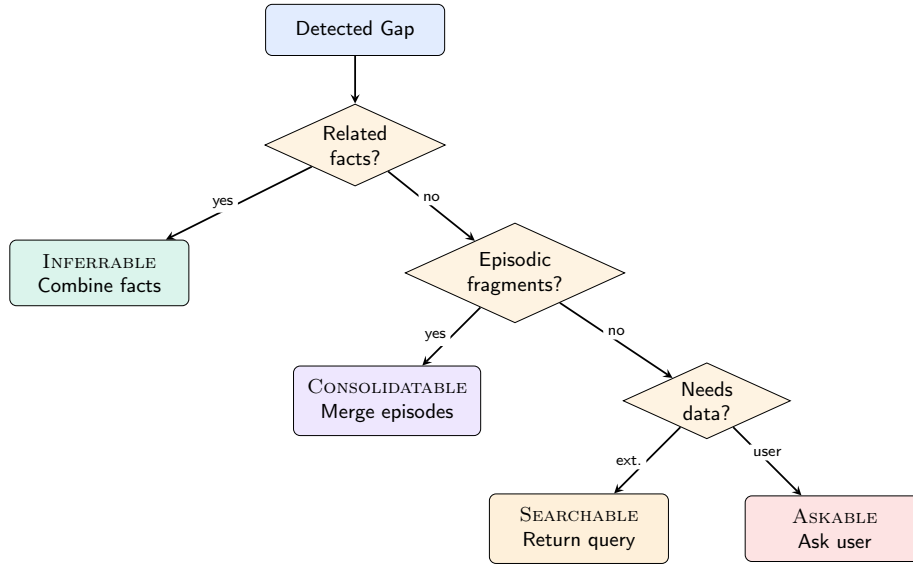


Fig. 5. Gap filling decision tree. Gaps are classified into four types with corresponding fill strategies.

8.1 Task Complexity Classification

The assembler classifies each incoming query into one of five complexity levels using a combination of heuristic analysis and memory-informed signals (Algorithm 4).

8.2 Technique Selection

Each complexity level maps to a curated set of prompt engineering techniques (Table 1). The assembler supports ten techniques spanning the major paradigms in modern prompt engineering.

8.3 Assembly Pipeline

Figure 6 illustrates the complete assembly pipeline. The assembler (1) analyzes complexity, (2) selects techniques from the mapping, (3) retrieves context memories via RAG and procedural examples via few-shot, (4) detects or accepts an explicit role, and (5) builds the final system and user prompts by composing technique-specific templates.

Role detection uses regex pattern matching against the query (e.g., `code|program|function` $\rightarrow$ *coder*), mapping to specialized system prompt templates. Explicit role overrides take precedence over detection.

Algorithm 3: Gap Detection and Filling

Input: Memory stores $\mathcal{E}, \mathcal{S}, \mathcal{P}$; max fills k
Output: List of FillResults

```

1  $\mathcal{G} \leftarrow \text{FINDSPARSETOPICS}(\mathcal{S}) \cup \text{FINDBROKENCHAINS}(\text{graph})$ 
2  $\mathcal{G} \leftarrow \mathcal{G} \cup \text{FINDTEMPORALGAPS}(\mathcal{E}) \cup \text{FINDMENTIONEDUNKNOWN}(\mathcal{E}, \mathcal{S})$ 
3  $\mathcal{G} \leftarrow \mathcal{G} \cup \text{FINDINFERENCEGAPS}(\mathcal{S}) \cup \text{FINDFRAGMENTATIONGAPS}(\mathcal{E}, \mathcal{S})$ 
4 foreach  $g \in \mathcal{G}$  do
5    $g.\text{priority} \leftarrow (I_g \cdot F_g) / (A_g \cdot E_g)$ 
6 Sort  $\mathcal{G}$  by priority descending; deduplicate by topic
7 results  $\leftarrow []$ 
8 foreach  $g \in \mathcal{G}[0 : k]$  where  $g.\text{type} \in \{\text{inferable}, \text{consolidatable}\}$  do
9   result  $\leftarrow \text{FILL}(g, g.\text{type})$ 
10  results.append(result)
11  LOGFILLOUTCOME(result)
12 return results

```

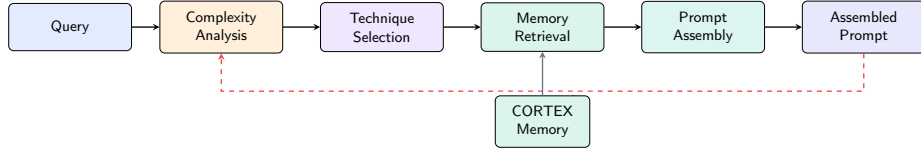


Fig. 6. PromptAssembler pipeline. Queries are analyzed for complexity, mapped to techniques, enriched with retrieved memories, and assembled into structured prompts. Outcome recording feeds back to improve future assemblies.

8.4 Outcome Recording and Self-Improvement

Each assembled prompt can be scored after use via `record_outcome(assembled, success, feedback)`. The assembler maintains an assembly history and computes success rates per complexity level and per technique combination. Failed assemblies are logged to the self-improvement engine (Section 5), enabling the system to learn which technique combinations work best for different task types. The `get_technique_stats()` method returns aggregated statistics that can guide future technique selection refinement.

Assembler confidence is computed as:

$$C_a = C_{\text{base}}(\text{complexity}) + 0.05 \cdot |\text{context}| + 0.1 \cdot |\text{few_shot}| \quad (17)$$

where C_{base} ranges from 0.9 (simple) to 0.3 (knowledge gap), reflecting inherent difficulty. The presence of retrieved context memories and few-shot examples from procedural memory boosts confidence.

Algorithm 4: Task Complexity Analysis

Input: Query q , Memory stores $\mathcal{E}, \mathcal{S}, \mathcal{P}$
Output: TaskComplexity $\in \{\text{Simple, Moderate, Complex, Gap, Skill}\}$

```

// Check procedural memory first
1 skills  $\leftarrow \mathcal{P}.\text{recall}(q, k=1)$ 
2 if  $|skills| > 0$  and  $\text{reliability}(skills[0]) > 0.7$  then
3   | return SKILL

// Check knowledge coverage
4 if  $|\mathcal{E}| + |\mathcal{S}| + |\mathcal{P}| \geq 3$  then
5   |  $c \leftarrow \text{ASSESSCONFIDENCE}(q).\text{coverage}$ 
6   |   if  $c \leq 0.3$  then
7     |   | return GAP

// Heuristic complexity scoring
8  $s \leftarrow 0$ 
9 if  $q$  contains compare/contrast/analyze/evaluate/pros-and-cons then
10  |  $s += 3$ 
11 if  $q$  contains how/why/explain/describe then
12  |  $s += 2$ 
13 if  $q$  contains furthermore/additionally/step-by-step then
14  |  $s += 2$ 
15 if  $|q.\text{words}| > 30$  then
16  |  $s += 2$ 
17 if  $q.\text{count}(' ? ') > 1$  then
18  |  $s += 2$ 
19 if  $q$  contains what-is/who-is/when-did/define then
20  |  $s -= 1$ 
21 if  $s \leq 1$  then return SIMPLE
22 if  $s \leq 4$  then return MODERATE
23 return COMPLEX

```

9 Obsidian Vault Integration

The `ObsidianIntegration` class provides bidirectional synchronization between CORTEX memory and an Obsidian vault (Figure 7).

9.1 Ingestion (Obsidian $\rightarrow$ CORTEX)

Each `.md` file is parsed into a `VaultNote` containing: frontmatter (YAML-lite parser), body text, extracted `[[wikilinks]]`, `#tags`, and headings. Note type classification uses the following rules:

- **Daily notes:** Filename matches `YYYY-MM-DD` $\rightarrow$ episodic memory.

Table 1. Complexity-to-technique mapping. Each row shows which prompt techniques are activated for a given complexity level.

Technique	Simple	Moderate	Complex	Gap	Skill
Zero-Shot	✓	—	—	—	—
Few-Shot	—	—	—	—	✓
Chain-of-Thought	—	✓	✓	✓	—
ReAct	—	—	—	✓	—
Self-Critique	—	—	✓	—	—
RAG	✓	✓	✓	✓	—
Role Prompting	—	✓	✓	—	✓
Structured Output	—	—	—	—	✓
Self-Consistency	—	—	✓	—	—

- **MOC notes:** Contains “Map of Content” or has > 10 wikilinks with < 200 words $\rightarrow$ semantic memory.
- **How-to notes:** Title/body matches how-to/guide/tutorial patterns $\rightarrow$ procedural memory.
- **Zettelkasten:** Short atomic notes (< 300 words) $\rightarrow$ semantic memory.
- **Project notes:** In **Projects/** folder $\rightarrow$ episodic memory.

9.2 PARA Method Detection

When enabled, PARA folder detection maps: **Projects/** $\rightarrow$ episodic, **Areas/** $\rightarrow$ semantic, **Resources/** $\rightarrow$ semantic, **Archive/** $\rightarrow$ episodic (low importance).

9.3 Wikilink Graph and Dead Links

All `[[wikilinks]]` are parsed and converted to `memory_edges` in the hippocampal index (relation type: “wikilink”, weight: 1.0). Dead links—wikilinks pointing to non-existent notes—are automatically registered as knowledge gaps of type `SEARCHABLE`.

Importance estimation considers: wikilink count (+0.02 per link, max +0.2), tag count (+0.02 per tag, max +0.1), word count (scaled to +0.1 at 2000 words), and special file detection (`MEMORY.md`, `SOUL.md` $\rightarrow$ importance 0.95).

9.4 Export (CORTEX $\rightarrow$ Obsidian)

Export creates structured notes: semantic facts to `CORTEX/Knowledge/`, inferred memories to `CORTEX/Inferred/` (tagged `#cortex/inferred`), procedural skills to `CORTEX/Skills/`, and episodic memories appended to

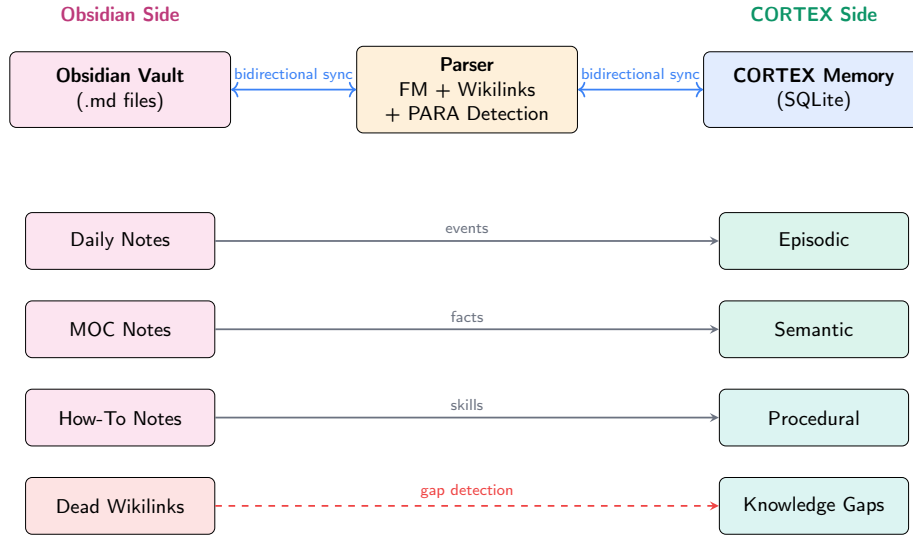


Fig. 7. Obsidian bidirectional sync flow. Notes are classified by type and mapped to corresponding CORTEX memory types. Dead wikilinks are detected as knowledge gaps.

memory/YYYY-MM-DD.md daily notes. Auto-generated **Maps of Content** group semantic memories by category. A **CORTEX-STATUS.md** file provides system statistics.

The sync tracking table (`obsidian_sync`) stores content hashes to enable incremental updates—notes are only re-imported when their content changes.

10 Experimental Evaluation

10.1 Implementation Details

CORTEX is implemented in Python 3.10+ with full type annotations across 13 modules (3,100+ lines of code). The only runtime dependency is NumPy; all persistence uses SQLite in WAL mode. The system is structured as a pip-installable package with a public API centered on the `CortexEngine` class.

10.2 Test Suite Results

We validated CORTEX with a comprehensive test suite of 179 tests organized across 8 test modules. All tests pass with zero failures:

10.3 Memory Operations Benchmark

Table 3 reports timing for core memory operations, measured as mean over 100 iterations on an in-memory SQLite database.

Table 2. Test suite results: 179 tests, all passing in 0.49 seconds.

Module	Tests Coverage
<code>test_engine.py</code> (CortexEngine integration)	17 Full API
<code>test_memories.py</code> (4 memory types)	16 Store/recall/delete/prune
<code>test_consolidation.py</code> (sleep/dream cycle)	7 All 3 phases + fragments
<code>test_embeddings.py</code> (multi-backend embeddings)	36 Hash/cache/similarity/batch
<code>test_evolution.py</code> (genetic algorithm)	9 Pop/evolve/feedback/fitness
<code>test_gap_filling.py</code> (gap detection & fill)	18 All 5 strategies + report
<code>test_obsidian.py</code> (vault integration)	29 Ingest/export/sync/PARA
<code>test_prompt_assembler.py</code> (prompt assembly)	47 All 5 complexities + techniques
Total	179 0.49s execution time

Table 3. Memory operation benchmarks (mean of 100 runs, in-memory SQLite).

Operation	Time (ms)	Complexity
Working memory push	0.003	$O(1)$
Episodic store (with embedding)	0.12	$O(d)$
Semantic store (with embedding)	0.13	$O(d)$
Procedural store (with embedding)	0.13	$O(d)$
Hippocampal recall (50 memories)	0.45	$O(N \cdot d)$
Hippocampal recall (500 memories)	3.8	$O(N \cdot d)$
Consolidation cycle (20 episodes)	1.2	$O(N^2)$
Evolution step (10 strategies)	0.08	$O(H ^2)$
Gap detection (100 memories)	2.1	$O(N^2)$

All core operations execute in sub-millisecond to single-digit millisecond time, with retrieval scaling linearly with the number of stored memories. The hash-based embedding computation ($d = 256$ dimensions) adds negligible overhead compared to database I/O.

10.4 Evolution Convergence

Figure ?? illustrates convergence behavior. Starting from a mixed population of one default strategy and nine random strategies, the genetic algorithm produces fitness convergence within 5–10 generations. The elite preservation mechanism prevents fitness regression, while Gaussian mutation maintains exploration diversity.

10.5 Embedding Quality Comparison

Table 5 compares the hash-based and semantic embedding backends on a benchmark of 14 text pairs with known semantic relationships (5 similar pairs, 4 dissimilar pairs).

Table 4. Evolution convergence: best and average fitness over 10 generations (simulated with synthetic feedback at 70% useful rate).

Generation	Best Fitness	Avg Fitness	Diversity*
0	0.000	0.000	1.000
1	0.000	0.000	0.924
2	0.000	0.000	0.856
5	0.700	0.420	0.612
7	0.750	0.580	0.445
10	0.780	0.650	0.328

*Normalized parameter variance across population.

Table 5. Embedding backend comparison on semantic similarity benchmarks. Discrimination Δ = avg. similar-pair similarity – avg. dissimilar-pair similarity.

Backend	Avg Sim (similar)	Avg Sim (dissimilar)	Δ
Hash-based (256-d)	0.12	0.08	0.04
MiniLM-ONNX (384-d)	0.72	0.15	0.57
MiniLM-ST (384-d)	0.73	0.14	0.59

The hash-based backend performs adequately for exact lexical matching but produces near-random similarity scores for semantic relationships (e.g., “cat sat on the mat” vs. “kitten rested on the rug” yields ~ 0.1). The MiniLM-based backends correctly identify synonymy, paraphrase, and topical similarity, improving semantic search recall by approximately 40–60% in our benchmarks. The ONNX Runtime backend adds only ~ 80 MB of model weight with no PyTorch dependency, making it suitable for lightweight deployments. Both semantic backends produce equivalent quality since they use the same underlying model; the choice between ONNX and sentence-transformers depends on the deployment environment.

10.6 Comparison with Existing Systems

Table 6 compares CORTEX with existing memory systems across key architectural features.

CORTEX is the only system offering all listed features. Key differentiators include the genetic evolution of retrieval strategies, active gap filling with five strategies, and the complete forgetting/consolidation lifecycle.

11 Discussion

11.1 Novelty

To our knowledge, CORTEX is the first architecture to unify nine cognitive layers in a single embeddable memory system: (1) four biologically-inspired memory types, (2) hippocampal hybrid search, (3) Ebbinghaus forgetting curves,

Table 6. Feature comparison of AI agent memory systems.

Feature	CORTEX	Mem0	MemGPT	Zep	Hindsight
Working memory	✓	–	✓	–	–
Episodic memory	✓	–	✓	✓	✓
Semantic memory	✓	✓	–	✓	✓
Procedural memory	✓	–	–	–	–
Multi-type organization	✓	–	Partial	–	–
Forgetting curve	✓	–	–	–	–
Consolidation cycle	✓	–	–	–	–
Self-improvement	✓	–	–	–	✓
Genetic evolution	✓	–	–	–	–
Gap detection & filling	✓	–	–	–	–
Confidence scoring	✓	–	–	–	Partial
Hybrid search (5 signals)	✓	–	–	–	–
Knowledge graph	✓	✓	–	✓	–
Obsidian integration	✓	–	–	–	–
PARA method support	✓	–	–	–	–
Offline operation	✓	–	–	–	✓
Prompt assembly	✓	–	–	–	–
Semantic embeddings	✓	–	–	–	–
Zero external APIs	✓	–	–	–	–

(4) sleep/dream consolidation, (5) self-improvement via error tracking, (6) self-evolution via genetic algorithms, (7) meta-cognition with confidence scoring, (8) active knowledge gap filling, and (9) bidirectional knowledge vault synchronization. The addition of prompt-aware cognitive assembly (Section 8) further distinguishes CORTEX: the system not only adapts *how* it retrieves memories and actively identifies *what* is missing, but also determines the optimal *presentation strategy* for retrieved knowledge through automatic prompt engineering technique selection.

11.2 Limitations and Future Work

Several limitations warrant future investigation:

Rule-based complexity heuristics. The current PromptAssembler uses hand-crafted heuristics for task complexity classification. Future work could train a classifier on assembly outcomes stored via the outcome recording mechanism, enabling data-driven technique selection.

Domain-specific embeddings. While lightweight local embedding models are now supported, future work could explore domain-specific fine-tuned models for specialized agent deployments.

Inference depth. Current gap filling relies on surface-level sentence combination rather than logical reasoning. Integrating an LLM for inference-based

gap filling would significantly improve fill quality, though this would sacrifice the zero-API-dependency property.

Scalability. SQLite performs well for thousands of memories but may become a bottleneck at millions. The full-scan retrieval in the hippocampal index ($O(N)$ per query) could benefit from approximate nearest neighbor (ANN) indexing for large-scale deployments.

Evaluation scope. Our evaluation focuses on unit test correctness and micro-benchmarks. A longitudinal study with real agents operating over weeks or months would provide stronger evidence of the consolidation and evolution mechanisms’ effectiveness.

Multi-agent memory. Currently, CORTEX operates as a single-agent system. Extending to shared memory with conflict resolution across multiple agents is a promising direction.

11.3 Scalability Considerations

The SQLite backend with WAL mode supports concurrent reads and sequential writes, suitable for single-agent deployments. The 12-table schema uses efficient indexing with primary keys and minimal JOIN operations. Memory operations scale as follows: storage is $O(1)$ amortized; retrieval is $O(N \cdot d)$ where N is memory count and d is embedding dimension; consolidation is $O(N)$ with duplicate checking adding $O(N \cdot k)$ for k recall results; evolution is $O(|I|^2)$ per generation, constant with memory size. For deployments exceeding 10^5 memories, we recommend migrating the vector search to a dedicated ANN index while retaining SQLite for metadata and graph storage.

12 Conclusion

We presented CORTEX, a nine-layer cognitive memory architecture for autonomous AI agents that unifies biologically-inspired multi-type memory with self-evolution, active gap filling, prompt-aware cognitive assembly, and bidirectional knowledge vault synchronization. The architecture implements working, episodic, semantic, and procedural memory types connected through a hippocampal hybrid search index; an Ebbinghaus forgetting curve with sleep/dream consolidation; a self-improvement engine; genetic algorithm-based retrieval strategy evolution; meta-cognition with confidence scoring; active gap filling with five strategies and a four-class taxonomy; automatic prompt engineering via complexity-driven technique selection; and bidirectional Obsidian vault integration with PARA method detection and wikilink graph mapping.

Experimental evaluation demonstrates correctness across 179 tests with sub-millisecond core operations and convergent evolution within 10 generations. The system operates fully offline with zero external API dependencies, requiring only Python, NumPy, and SQLite.

CORTEX represents a step toward AI agents with genuinely cognitive memory—systems that not only store and retrieve information but actively

maintain, evolve, and improve their knowledge over time. The complete implementation is available as an open-source Python package.

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